

SSIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – DATA CLEANING CHALLENGE

Course Code: DSA0502

CO Assessed: CO3

Weightage:

Course Title: Query Processing for Data Science

Assessment: Tool 2 – Data Cleaning Challenge

Total Marks:

CO3 Statement: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex. (BL5 and BL6).

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Section / Batch: 2024

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Project Mode: Individual Submission

Code of Conduct

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature:Sandrasudevan

Anomaly Detection (5 Marks)	Cleaning Methods (5 Marks)	Data Transformation (5 Marks)	Analytical Comparison (5 Marks)	Governance Discussion (5 Marks)	Total (25 Marks)

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs

```
mysql> INSERT INTO CustomerTransactions VALUES
-> ('C101','Raj Kumar','Male',28,'Electronics',2,'15000 INR','10-01-2025','Chennai'),
-> ('C102','Priya S','Female',35,'Fashion',3,'1200 INR','11-01-2025','chennai'),
-> ('C103','Arun V','Male',150,'Electronics',1,'300 USD','12-01-2025','CHENNAI'),
-> ('C104','Kavya R','Female',0,'Grocery',-2,'800 INR','15-01-2025','Coimbatore'),
-> ('C105','Rahul P','Male',42,'Fashion',5,'2500 INR','18-01-2025','Madurai'),
-> ('C105','Rahul P','Male',42,'Fashion',5,'2500 INR','18-01-2025','Madurai'),
-> (NULL,'Sneha M','Female',30,'Electronics',2,'450 USD','05-02-2025','Trichy'),
-> ('C107','Deepak N','Male',NULL,'Grocery',3,'600 INR','07-02-2025','Tiruchirappalli'),
-> ('C108','Meena K','Female',27,'Fashion',1,'1800 INR','2025-13-01','Salem'),
-> ('C109','John D','Male',31,'Electronics',4,'250 EUR','15-02-2025','Chennai');
Query OK, 10 rows affected (0.634 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE CustomerID IS NULL;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| CustomerID | CustomerName | Gender | Age | ProductCategory | Quantity | UnitPrice | PurchaseDate | City |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| NULL      | Sneha M      | Female | 30  | Electronics     | 2        | 450 USD   | 05-02-2025   | Trichy |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.070 sec)
```

- Duplicate transactions

```
mysql> SELECT CustomerID, CustomerName, COUNT(*) AS Count
-> FROM CustomerTransactions
-> GROUP BY CustomerID, CustomerName
-> HAVING COUNT(*) > 1;
```

CustomerID	CustomerName	Count
C105	Rahul P	2

1 row in set (0.198 sec)

- Negative quantities

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE Quantity < 0;
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore

1 row in set (0.021 sec)

- Unit prices recorded in different currencies

```
mysql> SELECT DISTINCT
-> SUBSTRING_INDEX(UnitPrice, ' ', -1) AS Currency
-> FROM CustomerTransactions;
```

Currency
INR
USD
EUR

3 rows in set (0.079 sec)

- Age values such as 0, 150, and blanks

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE Age <= 0 OR Age > 100 OR Age IS NULL;
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C107	Deepak N	Male	NULL	Grocery	3	600 INR	07-02-2025	Tiruchirappalli

3 rows in set (0.046 sec)

- Inconsistent city names (e.g., Chennai, Chennai, CHENNAI)

```
mysql> SELECT DISTINCT City
-> FROM CustomerTransactions;
```

City
Chennai
Coimbatore
Madurai
Trichy
Tiruchirappalli
Salem

6 rows in set (0.039 sec)

- Incorrect purchase dates

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE PurchaseDate = '2025-13-01';
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem

1 row in set (0.032 sec)

Tasks

1. Identify and classify all data quality issues. (5 Marks)

Missing Customer ID

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE CustomerID IS NULL;
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy

1 row in set (0.070 sec)

Duplicate Transaction

```
mysql> SELECT CustomerID, CustomerName, COUNT(*) AS Count
-> FROM CustomerTransactions
-> GROUP BY CustomerID, CustomerName
-> HAVING COUNT(*) > 1;
```

CustomerID	CustomerName	Count
C105	Rahul P	2

1 row in set (0.198 sec)

Negative Quantity

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE Quantity < 0;
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore

1 row in set (0.021 sec)

Invalid/Missing Age

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE Age <= 0 OR Age > 100 OR Age IS NULL;
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C107	Deepak N	Male	NULL	Grocery	3	600 INR	07-02-2025	Tiruchirappalli

3 rows in set (0.046 sec)

Different Currencies

```
mysql> SELECT DISTINCT
-> SUBSTRING_INDEX(UnitPrice, ' ', -1) AS Currency
-> FROM CustomerTransactions;
```

Currency
INR
USD
EUR

3 rows in set (0.079 sec)

Inconsistent Cities

```
mysql> SELECT DISTINCT City
-> FROM CustomerTransactions;
```

City
Chennai
Coimbatore
Madurai
Trichy
Tiruchirappalli
Salem

6 rows in set (0.039 sec)

Incorrect Date

```
mysql> SELECT *
-> FROM CustomerTransactions
-> WHERE PurchaseDate = '2025-13-01';
```

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem

```
1 row in set (0.032 sec)
```

2. Design a cleaning strategy for each issue. (5 Marks)

Fill missing Customer ID

```
mysql> UPDATE CustomerTransactions
-> SET CustomerID = 'C106'
-> WHERE CustomerID IS NULL;
Query OK, 0 rows affected (1.372 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
mysql> SELECT CustomerID, CustomerName
-> FROM CustomerTransactions
-> WHERE CustomerName = 'Sneha M';
```

CustomerID	CustomerName
C106	Sneha M

```
1 row in set (0.409 sec)
```

Remove duplicate

```
mysql> UPDATE CleanedTransactions
      -> SET City = 'Chennai'
      -> WHERE LOWER(City) = 'chennai';
Query OK, 2 rows affected (0.195 sec)
Rows matched: 4  Changed: 2  Warnings: 0

mysql> UPDATE CleanedTransactions
      -> SET City = 'Tiruchirappalli'
      -> WHERE City = 'Trichy';
Query OK, 1 row affected (0.069 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE CleanedTransactions
      -> SET Quantity = NULL
      -> WHERE Quantity < 0;
Query OK, 1 row affected (0.064 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE CleanedTransactions
      -> SET Age = NULL
      -> WHERE Age <= 0 OR Age > 100;
Query OK, 2 rows affected (0.067 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> UPDATE CleanedTransactions
      -> SET PurchaseDate = NULL
      -> WHERE PurchaseDate = '2025-13-01';
Query OK, 1 row affected (0.052 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

3. Perform data standardization and transformation. (5 Marks)

```
mysql> SELECT CustomerName, PriceAmount, Currency, PriceINR
-> FROM CleanedTransactions;
```

CustomerName	PriceAmount	Currency	PriceINR
Raj Kumar	15000.00	INR	15000.00
Priya S	1200.00	INR	1200.00
Arun V	300.00	USD	25500.00
Kavya R	800.00	INR	800.00
Sneha M	450.00	USD	38250.00
Deepak N	600.00	INR	600.00
Meena K	1800.00	INR	1800.00
John D	250.00	EUR	22500.00

8 rows in set (0.010 sec)

4. Analyze how cleaning affects total revenue calculations. (5 Marks)

```
mysql> SELECT
-> CustomerName,
-> ProductCategory,
-> Quantity,
-> PriceINR,
-> Quantity * PriceINR AS Revenue
-> FROM CleanedTransactions;
```

CustomerName	ProductCategory	Quantity	PriceINR	Revenue
Raj Kumar	Electronics	2	15000.00	30000.00
Priya S	Fashion	3	1200.00	3600.00
Arun V	Electronics	1	25500.00	25500.00
Kavya R	Grocery	NULL	800.00	NULL
Sneha M	Electronics	2	38250.00	76500.00
Deepak N	Grocery	3	600.00	1800.00
Meena K	Fashion	1	1800.00	1800.00
John D	Electronics	4	22500.00	90000.00

8 rows in set (0.025 sec)

```
mysql> SELECT
-> SUM(Quantity * PriceINR) AS TotalRevenue
-> FROM CleanedTransactions
-> WHERE Quantity > 0;
```

TotalRevenue
229200.00

1 row in set (0.107 sec)

5. Generate a report discussing business risks if the data were not cleaned. (5 Marks)

If the company uses the raw data without cleaning:

1. **Incorrect revenue reports** – Duplicate transactions can artificially increase sales revenue.
2. **Wrong product performance** – Incorrect quantities and prices can make some products appear more or less successful.
3. **Poor customer segmentation** – Invalid or missing customer information can affect customer analysis.
4. **Incorrect regional analysis** – Chennai, chennai, and CHENNAI may be treated as different cities.
5. **Incorrect financial decisions** – Mixed currencies can produce highly inaccurate revenue calculations.
6. **Inventory problems** – Negative or incorrect quantities can cause incorrect stock levels.
7. **Poor management decisions** – Business strategies based on inaccurate reports may result in wrong marketing, pricing, and inventory decisions.

Conclusion

Data cleaning converts the raw transaction dataset into a **consistent, accurate, and reliable dataset**. This improves revenue calculations, customer segmentation, product analysis, inventory management, and overall business decision-making.

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues

- Missing credit scores

```
mysql> SELECT CustomerID, CreditScore
-> FROM CleanedLoanData
-> WHERE CreditScore IS NULL;
```

CustomerID	CreditScore
L003	NULL

1 row in set (0.013 sec)

- Income values recorded in different units

```
mysql> SELECT CustomerID, AnnualIncome
-> FROM LoanData
-> WHERE AnnualIncome LIKE '%K%';
```

CustomerID	AnnualIncome
L002	900K

1 row in set (0.115 sec)

- Negative loan amounts

```
mysql> SELECT CustomerID, LoanAmount
-> FROM LoanData
-> WHERE LoanAmount < 0;
```

CustomerID	LoanAmount
L004	-200000

1 row in set (1.622 sec)

- Duplicate customer IDs

```
mysql> SELECT CustomerID, COUNT(*) AS Count
-> FROM LoanData
-> GROUP BY CustomerID
-> HAVING COUNT(*) > 1;
+-----+-----+
| CustomerID | Count |
+-----+-----+
| L008       | 2     |
+-----+-----+
1 row in set (0.857 sec)
```

- Age values below 18

```
mysql> SELECT CustomerID, Age
-> FROM LoanData
-> WHERE Age < 18;
+-----+-----+
| CustomerID | Age  |
+-----+-----+
| L005       | 16   |
+-----+-----+
1 row in set (1.209 sec)
```

- Employment status entered differently (FT, Full Time, fulltime)

```
mysql> SELECT DISTINCT EmploymentStatus
-> FROM LoanData;
+-----+
| EmploymentStatus |
+-----+
| Full Time        |
| FT               |
| fulltime         |
| Part Time        |
| Self Employed    |
| Unemployed       |
+-----+
6 rows in set (0.068 sec)
```

- Extreme outliers in income

```
mysql> SELECT CustomerID, AnnualIncome
-> FROM LoanData
-> WHERE CAST(
->     REPLACE(AnnualIncome, 'K', '000')
->     AS UNSIGNED
-> ) > 10000000;
```

```
+-----+-----+
| CustomerID | AnnualIncome |
+-----+-----+
| L007       | 75000000    |
+-----+-----+
1 row in set (0.936 sec)
```

Tasks

1. Identify critical issues that could impact loan decisions. (5 Marks)

```
+-----+-----+
| Issue                                     |
+-----+-----+
| Missing Credit Score - L003             |
| Negative Loan Amount - L004             |
| Duplicate Customer ID - L008            |
| Age Below 18 - L005                     |
| Inconsistent Employment Status - FT/fulltime |
| Extreme Income Outlier - L007           |
+-----+-----+
6 rows in set (1.206 sec)
```

2. Propose cleaning techniques for missing and inconsistent data. (5 Marks)

Missing credit score

```
mysql> SELECT CustomerID, CreditScore
-> FROM CleanedLoanData
-> WHERE CreditScore IS NULL;
```

```
+-----+-----+
| CustomerID | CreditScore |
+-----+-----+
| L003       | NULL        |
+-----+-----+
1 row in set (0.013 sec)
```

Standardize income

```
mysql> SELECT CustomerID, AnnualIncome
-> FROM LoanData
-> WHERE CustomerID = 'L002';
```

CustomerID	AnnualIncome
L002	900K

```
1 row in set (0.481 sec)
```

Standardize employment

3. Detect and justify treatment of outliers. (5 Marks)

```
mysql> SELECT CustomerID, AnnualIncome
-> FROM LoanData
-> WHERE CAST(AnnualIncome AS UNSIGNED) > 10000000;
```

CustomerID	AnnualIncome
L007	75000000
L007	75000000

```
2 rows in set, 2 warnings (0.846 sec)
```

4. Analyze the effect of cleaning on loan approval rates. (5 Marks)

```
mysql> SELECT
-> COUNT(CASE WHEN LoanStatus = 'Approved' THEN 1 END) AS Approved,
-> COUNT(*) AS Total,
-> ROUND(
-> COUNT(CASE WHEN LoanStatus = 'Approved' THEN 1 END) * 100.0 / COUNT(*), 2
-> ) AS ApprovalRate
-> FROM LoanData;
```

Approved	Total	ApprovalRate
16	20	80.00

```
1 row in set (0.910 sec)
```

5. Discuss ethical implications of incorrect data cleaning decisions. (5 Marks)

```
mysql> SELECT
-> COUNT(CASE WHEN LoanStatus = 'Approved' THEN 1 END) AS Approved,
-> COUNT(*) AS Total,
-> ROUND(
-> COUNT(CASE WHEN LoanStatus = 'Approved' THEN 1 END) * 100.0 / COUNT(*), 2
-> ) AS ApprovalRate
-> FROM CleanedLoanData;
```

Approved	Total	ApprovalRate
7	9	77.78

1 row in set (0.550 sec)